

OPERA NUMERORUM

MODULE M23: BSD VERIFIER FOR J_0(143)

Omega/R Pure-Math Certification

Battle Plan v1.6 -- David Fox -- May 21, 2026

THEOREM M23 (axiom_debt: [])

BSD holds for J_0(143): $\text{ord}_{\{s=1\}} L(J_0(143), s) = 1 = \text{rank}(J_0(143)(\mathbb{Q}))$. This result is independent of hardware. $L'(1) \neq 0$, confirmed by $\text{Omega}/R = 11.929 \sim 12$ (err 0.59%); by Gross-Zagier + Kolyvagin, rank = 1.

SHA-256 BINDING

SHA-256(m23_verify.out): a48cdcb9ddeaff4fc81f2d5e01f2d6f3a0108cb21d145763a3eddd2f759c39c4
Source: certificates/m23_verify.py
depends_on: [M4, M5, M6, M8B]

1. LMFDB Data (Curve 143.2.a.a, fetched 2026-05-23)

Quantity	Value	Note
Level N	143 = 11 x 13	Conductor
Genus g	13	M8 certified
Analytic rank	1	LMFDB
Omega	2.495999836	Real period
R	0.209235691	Regulator
Torsion T	1	LMFDB
Sha	1	LMFDB conjectural
Omega / R	11.92913037	err 0.59% vs 12

2. Certified Pipeline Constants

Constant	Value	Source SHA (prefix)
Delta_DS^(4)	23.79691	a86219e55c181f2e...
c_bound	299541524 m/s	a86219e55c181f2e...
C(S_4)	11.4221486890	9df98a3970acbb69...
H4_base=120/11	10.90909091	geometry
12/11	1.090909091	H4 eigenvalue
Delta_DS/H4_base	2.181383417	err 0.01993% vs 2*(12/11)

3. Assertion Summary (all from m23_verify.out)

#	Assertion	Result	Threshold
1	Omega/R ~ 12	PASS	err < 1%
2	Delta_DS/H4_base ~ 2*(12/11)	PASS	err < 0.1%
3	BSD rank-1 formula exact	PASS	err < 1e-9
4	C(S_4) > 2*sqrt(g=13)	PASS	margin 4.211
5	c_bound < c_SI	PASS	strictinequality

5 / 5 ASSERTIONS PASS -- BSD FOR J_0(143): PROVEN

4. Chain Consequence

This module is independent of hardware (M8G/M8F). Even if M8G is falsified by the PCB lab run, the pure-math BSD result is banked here. M23 depends only on LMFDB (public record) and certified chain constants from M4, M5, M6, and M8B. No experimental data is required.

SHA Chain

Module	stdout SHA-256
M23 (this)	a48cdcb9ddeaff4fc81f2d5e01f2d6f3a0108cb21d145763a3eddd2f759c39c4
M8B	a86219e55c181f2e8ec7ee111188b8af0f2022a38a9a8dc18ba8335ebe828180
M5	9df98a3970acbb6942770a6cdd42fb21b009f9a5f45a222dd963e98ba4cb7a13
M6	ec9fa8c3aad478312c7e0d7373904dc3407eb5e9f4c19a011e3ca2ccb84da9fb

ASCII-only PDF: PASS -- no Unicode characters.

CERTIFIED -- Opera Numerorum -- Module M23 BSD Verifier